

Climate disasters and uncertain futures: Quantifying change in the environmental sciences

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UBC Postdoctoral Research Day, Dec. 2021

A fundamental challenge of Earth systems modelling

Example: **How is climate change impacting Amazonian wetlands?**



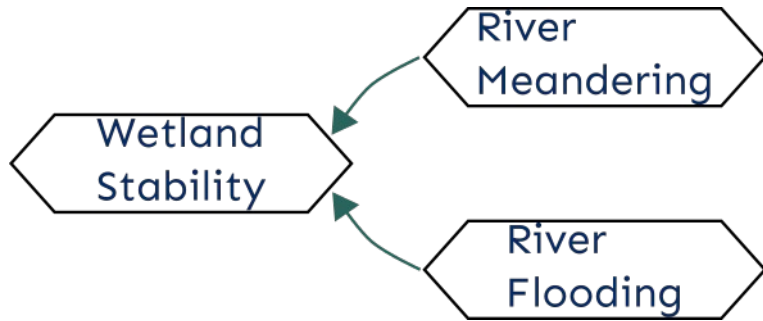
Wetland
Stability



In Earth science, system **boundaries** are **not well defined**

A fundamental challenge of Earth systems modelling

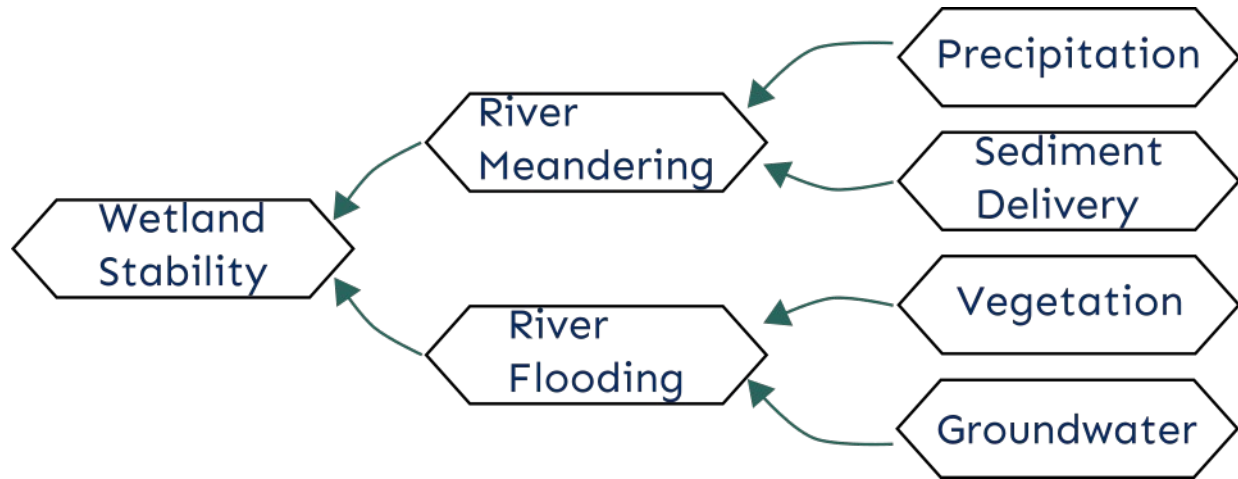
Example: **How is climate change impacting Amazonian wetlands?**



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A fundamental challenge of Earth systems modelling

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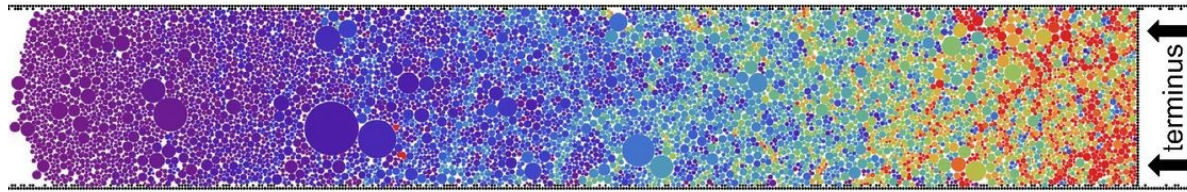
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Main approaches to model Earth systems

1. Computational

Quantifying flow and stress in ice mélange,
the world's largest granular material

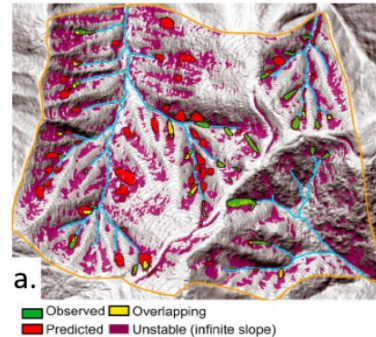
Justin C. Burton^{a,1}, Jason M. Amundson^b, Ryan Cassotto^c, Chin-Chang Kuo^d, and Michael Dennin^d



2. Data driven

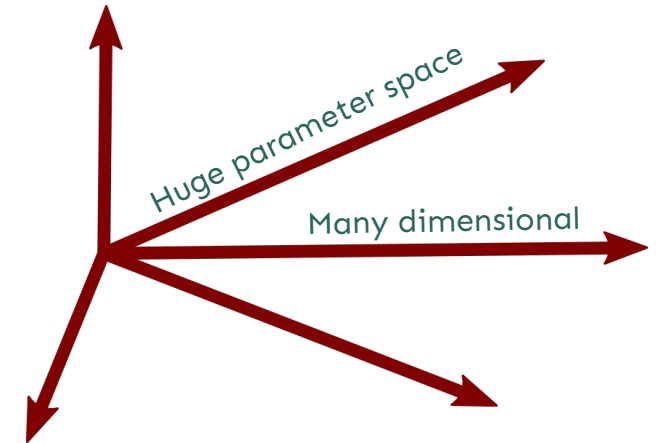
Predicting shallow landslide size and location across
a natural landscape: Application of a spectral
clustering search algorithm

Dino Bellugi¹, David G. Milledge², William E. Dietrich³, J. Taylor Perron¹, and Jim McKean⁴



CHALLENGES:

- ◆ Site specific
- ◆ Requires calibration
- ◆ **Difficult to generalize**



We need general conclusions

An alternate approach to model Earth systems

Stochastic methods:

Hans Einstein



Open system issue is side-stepped using **statistical mechanics**

Probabilistic models **admit analytical solutions** and **produce general conclusions**



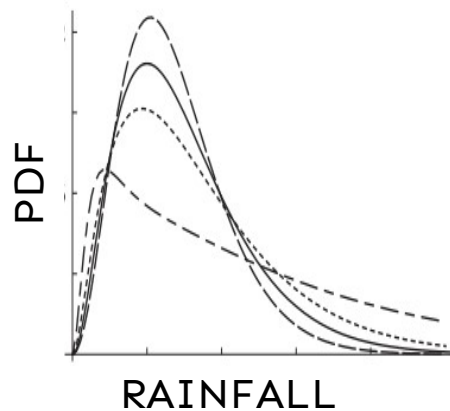
Resilience of river flow regimes

Gianluca Botter^a, Stefano Basso^{a,b}, Ignacio Rodriguez-Iturbe^c, and Andrea Rinaldo^{a,d,1}

PNAS | August 6, 2013 | vol. 110 | no. 32 | 12925–12930

River discharge links to **flooding, drought, ecology**

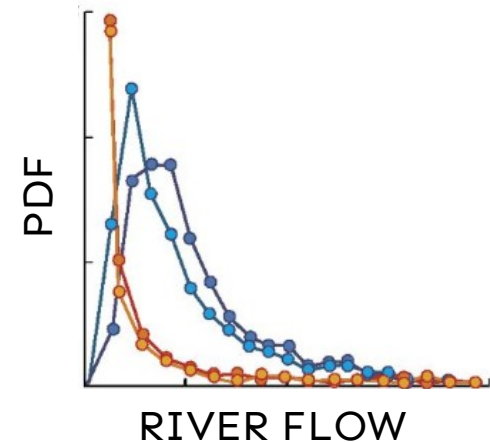
Input Distributions



Model



Output Distributions



General result: Rivers with less variable flows are more sensitive to climate change

Applications of stochastic methods: Soil salinity

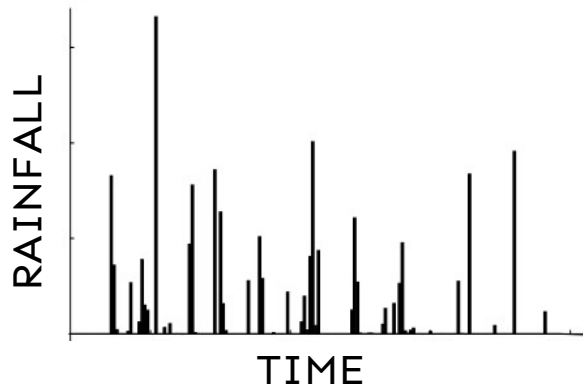


Stochastic modeling of soil salinity

S. Suweis,¹ A. Rinaldo,^{1,2} S. E. A. T. M. Van der Zee,³ E. Daly,^{4,5} A. Maritan,⁶ and A. Porporato^{1,7}

Soil salinity impacts **crop yields, drinking water**

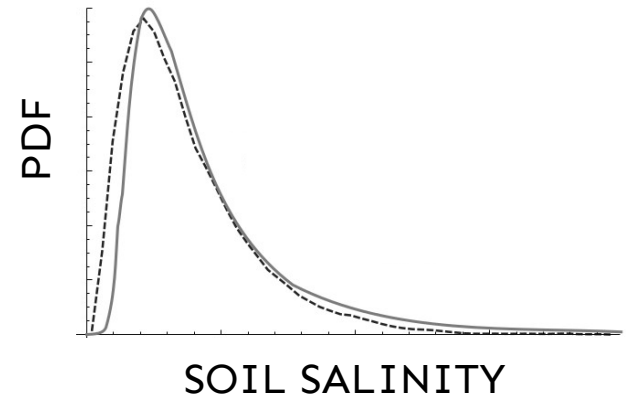
Input Distributions



Model



Output Distributions



General result: Soil salinity is extremely sensitive to climate change

Application of stochastic methods: Contaminant transport

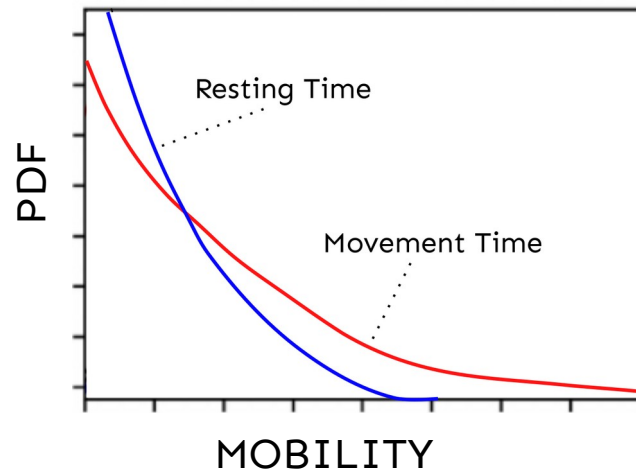


Back to Einstein: Burial-Induced Three-Range Diffusion in Fluvial Sediment Transport

J. Kevin Pierce¹  and Marwan A. Hassan¹ 

River contaminants inhibit aquatic organisms

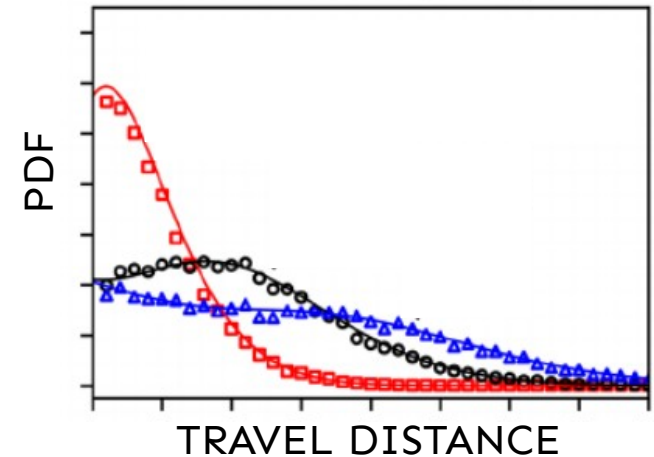
Input Distributions



Model



Output Distributions



General result: Contaminant diffusion becomes extremely slow due to burial

Application of stochastic methods: Channel Evolution



Stochastic description of the bedload sediment flux

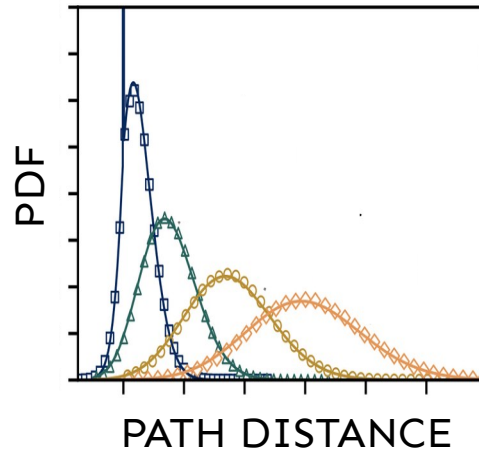
J. Kevin Pierce¹, Marwan A. Hassan¹, and Rui M.L. Ferreira²

¹The University of British Columbia, Vancouver, Canada

²Instituto Superior Técnico, Lisbon, Portugal

Sediment transport sets **channel evolution**

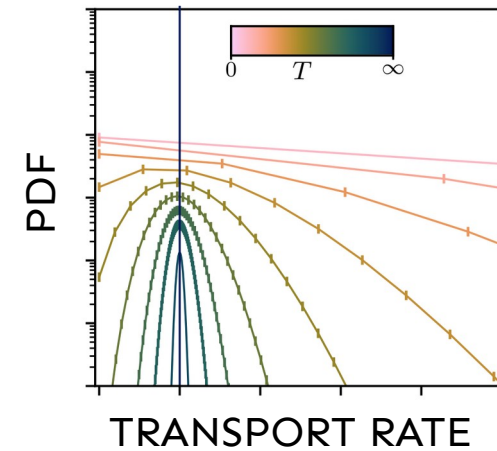
Input Distributions



Model



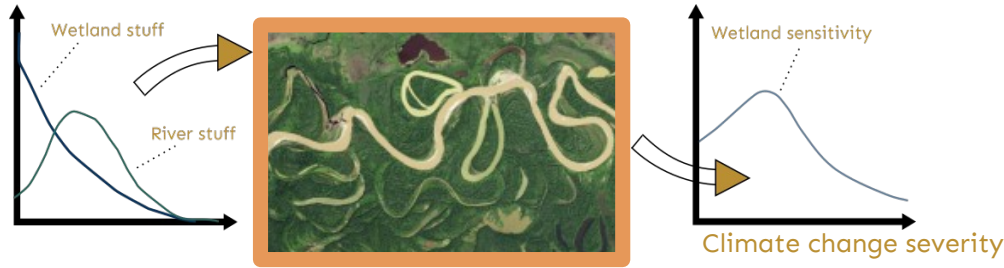
Output Distributions



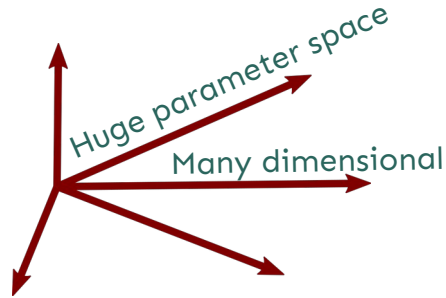
General result: Expected rates of sediment movement shift with the observation timescale

Summary: Probabilistic methods in Earth Science

1. Map input distributions to output distributions



2. Alternative to numerical or data-driven methods



Open systems

Analytical results

General conclusions

Uncertainty

3. Many further applications are possible

Permafrost, Methane, Landslides, Wildfires, ...



$n = 1$

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